

helixvpnctl — the operator CLI (Cobra)

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Master technical specification — Volume 6 (Deployment, Tooling & Operations), nano-detail document. Scope: the **helixvpnctl operator CLI** — the Cobra command binary that is both the homelab front door (`init + systemctl start` = a running gateway) and the GitOps front door (`policy set/policy compile`). This deepens [05-overview §5] into a full command-by-command spec: the command tree, per-command flags / inputs / outputs / exit codes, how each command talks to `svc-api` over device-or-operator mTLS, credentials handling (§11.4.10), and concrete `--help` sketches per command. It is a SPEC: command surfaces and skeletons are illustrative of the contract, not the shipping implementation (2–3 refinement passes follow).

Evidence cited inline by id: [05_OV §N] = 05-repo-layout-tooling-and-helix-ecosystem.md; [API §N] = v03-control-plane/svc-api.md; [PROTO §N] = v03-control-plane/protobuf-spec.md; [04_P1 §9] = HelixVPN-Phase1-MVP.md (helixvpnctl

- quadlets); [04_ARCH §4.7] = the operator-CLI-over-bash-scripts decision. Unproven facts are marked **UNVERIFIED** per §11.4.6 — never fabricated.
-

```
helixvpncctl
├─ init                --domain --data-dir [--substrate quadlet|
compose[k8s]          # $3
├─
keys
# gateway WG key ops ($4)
|   └─ rotate          [--
confirm]                #   rotate
gateway WG keypair
├─ └─
```

```

show
#   print gateway WG public key
├─ enroll-token      --kind client|connector [--site NAME] [--ttl
DUR] [--qr] # §5 mint enrollment token
├─
policy
# §6
|   └─ get           [--version N] [-o json|
jsonc]                #   fetch active/Nth policy
|   └─ set           ./policy.jsonc [--
activate]              #   submit (dry-run compile)
[+ activate]
|   └─ compile       ./
policy.jsonc                #   dry-run
compile ONLY (no persist)
├─
device
# §7
|   └─ list           [--kind ...] [--online] [-o table|
json]                  #   list tenant devices
|   └─ revoke        <id> [--
confirm]                #   revoke device
(<1s edge enforce)
├─
connector
# §8
|   └─ list           [-o table|
json]                  #   list connectors
+ advertised CIDRs
|   └─ advertise      <id> --cidr
CIDR[,CIDR...]          #   set advertised
CIDRs (declarative)
├─ status             [-o table|
json]                  # §9 edge health +
transport ladder
├─
deploy
# §10 (delegates to substrate docs)
├─ quadlet            --out DIR
├─ compose            --out DIR
├─ kube               --out DIR

```

This refines [05_OV §5.1]: policy apply→policy set, policy diff→policy compile, gateway keys→keys rotate/keys show, gateway status→status, network advertise→connector advertise, and join is deferred to Phase 2 (fleet member join) — recorded as a **visible reconciliation** (§11.4.35) so the two docs do not silently disagree on verb naming.

3. init — bootstrap a single-node gateway

```
$ helixvpncctl init --help
```

Bootstrap a single-node HelixVPN gateway: tenant, admin, CA, gateway WG keys, infra credentials, and the deploy units for the chosen substrate.

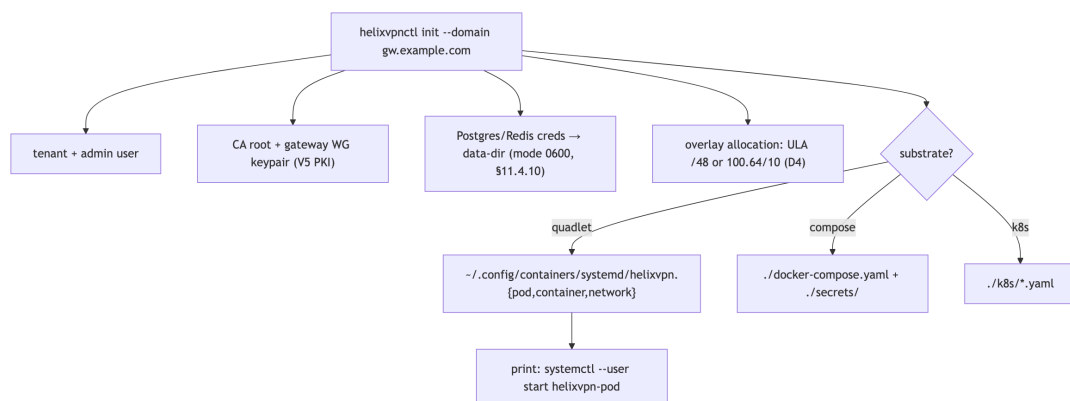
Usage:

```
helixvpncctl init --domain <FQDN> [flags]
```

Flags:

<code>--domain string</code>	public gateway FQDN (REQUIRED), e.g. gw.example.com
<code>--data-dir string</code>	state + secrets dir (default <code>"/var/lib/helixvpn"</code>)
<code>--substrate string</code>	quadlet compose k8s (default <code>"quadlet"</code>) # §11.4.76/.161 canonical=quadlet
<code>--overlay string</code>	ula cgnat overlay mode (default <code>"ula"</code>) # D4: ULA /48 vs 100.64/10
<code>--force</code>	re-run on a non-empty data-dir (backs up first, §9.2)
<code>-h, --help</code>	help for init

What init writes (the artifact set [05_OV §5.3, 04_P1 §9]):



Skeleton [05_OV §5.2]:

```
// helix-go/cmd/helixvpncctl/init.go (illustrative)
func initCmd() *cobra.Command {
    var domain, dataDir, substrate, overlay string
    var force bool
    c := &cobra.Command{Use: "init", Short: "Bootstrap a single-
node gateway",
    RunE: func(cmd *cobra.Command, _ []string) error {
        // 1. tenant + admin user + overlay /48 (or 100.64/10
        // - D4)
        // 2. CA root + first gateway WG keypair (V5 PKI)
        // 3. Postgres+Redis creds → data-dir (mode 0600,
        // §11.4.10)
        // 4. render deploy units for the chosen substrate
        // (default quadlet, §11.4.76)
```

```

        // 5. print: systemctl --user start helixvpn-pod
        return bootstrap(domain, dataDir, substrate, overlay,
            force)
    }}
    c.Flags().StringVar(&domain, "domain", "", "public gateway
        FQDN (required)")
    c.Flags().StringVar(&dataDir, "data-dir", "/var/lib/
        helixvpn", "state + secrets dir")
    c.Flags().StringVar(&substrate, "substrate", "quadlet",
        "quadlet|compose|k8s")
    c.Flags().StringVar(&overlay, "overlay", "ula", "ula|cgnat")
    c.Flags().BoolVar(&force, "force", false,
        "re-run on non-empty data-dir (backs up first)")
    _ = c.MarkFlagRequired("domain")
    return c
}

```

Exit codes: 0 success; 2 invalid flags (missing --domain, bad substrate); 3 data-dir non-empty without --force; 4 overlay allocation failed; 5 PKI/keygen failed; 6 substrate render failed. **UNVERIFIED:** the exact numeric mapping above is a design default; it is fixed at implementation time and surfaced in --help then.

4. keys — gateway WG key operations

\$ helixvpncctl keys rotate --help
 Rotate the gateway WireGuard keypair. The new public key is pushed to the control plane so the coordinator distributes it to every agent via WatchNetworkMap; old key is retired after a grace window so in-flight tunnels re-handshake without a hard drop.

Usage:
 helixvpncctl keys rotate [--confirm]

Flags:
 --confirm required for a destructive rotation (re-handshakes all agents)

- keys rotate is **destructive-ish** (every agent re-handshakes); it requires --confirm, and the rotation is reversible within the grace window (§11.4.101 — reversible + bounded blast radius → the CLI may proceed once confirmed, but never silently). Old key retained for a grace window (**UNVERIFIED:** default grace = the gateway's WG rekey interval; pinned in V5 PKI).
 - keys show prints the current gateway WG **public** key (never the private key — C6; the private key never leaves the gateway). Exit 0/2.
-

5. enroll-token — mint a device enrollment token

```
$ helixvpctl enroll-token --help
```

Mint a single-use, short-lived device enrollment token. The raw token is printed ONCE (only its sha256 is persisted server-side, §11.4.10); re-fetching shows metadata only.

Usage:

```
helixvpctl enroll-token --kind <client|connector> [flags]
```

Flags:

```
--kind string    client|connector (REQUIRED)
--site string    site label for a connector (optional)
--ttl duration   token lifetime (default 15m, min 1m, max 24h)
--qr             also print a QR PNG (base64) for Access/Connector scan
--bind-user string bind token to a user (omit => anonymous enroll, C6 privacy)
```

Maps to POST /v1/enroll-tokens [API §4.1]. Output (table form):

ID	2f9a...	Kind	connector
Token	hvpn_e_3K9x...	(shown ONCE – store it now)	
Expires	2026-06-25T13:15:00Z		
QR	[PNG written to ./enroll-2f9a.png] (with --qr)		

–o json emits {id, token, kind, expires_at, qr_png_b64}. **Anti-bluff (§11.4.10):** the raw token is the only place the secret ever appears; the CLI never writes it to a log, never echoes it under –o json to a file without 0600 perms, and a subsequent enroll-token list (metadata-only) cannot recover it. Exit 0; 2 bad flags; 5 mint failed (server 4xx/5xx).

6. policy — GitOps policy front door

```
$ helixvpctl policy set --help
```

Submit a policy spec. ALWAYS dry-run-compiles first (fail-closed): a compile error prints the structured PolicyCompileError and writes NOTHING. A clean compile persists a new policy version. --activate also flips it active (atomic), triggering per-agent deltas within the <1s convergence SLO.

Usage:

```
helixvpctl policy set <FILE.jsonc> [--activate]
```

Flags:

--activate activate the new version after a clean compile

Subcommand	REST call [API §4.5]	Behavior
policy compile FILE	POST /v1/policies (dry-run path)	compile ONLY; print CompileStats + any RouteConflict; persist nothing
policy set FILE	POST /v1/policies	dry-run compile → on clean, persist a new version (inactive)
policy set FILE --activate	POST /v1/policies then POST /v1/ policies/:v/ activate	persist + atomic activate (emits policy.compiled)
policy get [-- version N]	GET /v1/policies	fetch active (or Nth) policy as jsonc

Fail-closed evidence: a bad spec exits non-zero and prints the
PolicyCompileError (stage/field/reason/detail) from [API §10.1], e.g.:

```
$ helixvpctl policy set ./policy.jsonc
POLICY COMPILE FAILED (422):
  stage:  resolve_hosts
  field:  acls[1].dst[0]
  reason: cidr_not_advertised
  detail: 10.2.0.0/24 is not advertised by any connector
nothing written.
```

Exit 0 clean; 1 compile failed (policy_compile_failed, nothing written); 2
bad flags/file; 5 server/transport error. The dry-run-before-persist is the
load-bearing fail-closed guarantee [API §4.5] — the CLI never persists an
uncompilable policy.

7. device — device lifecycle

```
$ helixvpctl device revoke --help
Revoke a device. The control plane sets devices.revoked_at +
revokes the device cert,
emits device.revoked, force-closes the device's WatchNetworkMap
stream, and the edge
drops the WG peer – all within the <1s convergence SLO.
```

Usage:

```
helixvpctl device revoke <DEVICE_ID> [--confirm]
```

Flags:

 --confirm required (revocation is high-impact: the device loses access immediately)

Subcommand	REST call [API §4.2/ §4.3]	Behavior
device list	GET /v1/devices (cursor-paginated)	table: id, kind, name, overlay_ip, online, revoked
device revoke <id>	POST /v1/ devices/:id/revoke (admin only)	202 Accepted; CLI then polls/ streams for the < 1s enforcement confirmation

device list flags: --kind client|connector, --online, -o table|json, --limit (≤200), --cursor. The wg public key is **not** shown (need-to-know C4 — it is an agent-plane datum delivered via WatchNetworkMap, not exposed to the operator CLI) [API §4.2]. Exit 0; 2 bad flags; 3 --confirm missing on revoke; 4 device not found (404); 5 server error.

8. connector — connector advertised routes

```
$ helixvpncctl connector advertise --help
```

Set the COMPLETE set of CIDRs a connector serves (declarative, not a diff). Re-sending the same set is a no-op. Overlapping CIDRs across connectors are accepted with a conflict advisory (resolved via 4via6 site disambiguation, D4); a malformed CIDR is rejected.

Usage:

```
helixvpncctl connector advertise <CONNECTOR_ID> --cidr  
<CIDR>[,<CIDR>...]
```

Flags:

 --cidr strings complete declarative CIDR set (REQUIRED)

Maps to POST /v1/connectors/:id/prefixes (Console path) /

AdvertisePrefixes [PROTO §1, API §4]. connector list → GET /v1/connectors (table: device_id, site, prefixes, 4via6). The declarative-not-diff semantics are load-bearing [PROTO §4.5]: the CLI sends the **complete** set, the server diffs and emits

connector.prefixes.changed only on real change. A malformed CIDR exits 1 (prefix_conflict/validation_failed); an overlap exits 0 with a printed advisory.

9. status — gateway/edge health

```
$ helixvpncctl status
GATEWAY gw.example.com:443
  edge      UP      (helix-edge 1.0, readyz 200)
  control   UP      (helixd 1.0)
  postgres  UP      redis  UP
  overlay   fd7a:helix:91a3::/48  devices=7  connectors=2
  transport ladder: [plain-udp, lwo, masque-h3]  active-
default: plain-udp
```

Composes GET /readyz + GET /v1/networks + ReportStatus-derived presence [API §4.4/§4.7]. -o json emits the structured health object. Exit 0 all-up; 1 degraded (any component down — readyz 503); 5 unreachable (transport error). The transport-ladder posture (which obfuscation transports the gateway offers) is read-only here; the agent walks the ladder (Volume 2 transport-selection-ladder.md).

10. deploy — render substrate units (delegates to substrate docs)

```
$ helixvpncctl deploy quadlet --out ~/.config/containers/systemd/
```

Subcommand	Output	Substrate doc
deploy quadlet --out DIR	*.pod *.container *.network rootless units	podman- quadlets.md
deploy compose --out DIR	docker-compose.yaml + secrets/	docker- compose.md
deploy kube -- out DIR	*.yaml (DaemonSet/Deployment/ StatefulSet/Service/ NetworkPolicy)	kubernetes.md

deploy renders the three substrates from **one in-code spec** via the containers submodule's pkg/compose primitives (§11.4.76, §11.4.81 cross-platform-parity: one source, per-substrate render) [05_OV §6.3]. It never hand-maintains three divergent files. The unit/manifest content is specified in the substrate docs, not here.

11. Talking to svc-api — auth, transport, config

11.1 Transport

helixvpctl is a REST client of svc-api (the Gin surface [API §3]); on a freshly-inited single-node gateway it talks to `https://localhost:8443` over the loopback. Two auth modes:

Mode	Credential	Used for	Source
Operator API token	Authorization: Bearer hvpn_<...>	all REST ops (enroll- token, policy, device, connector, status)	init writes an admin API token to the data- dir; --token/ \$HELIXVPCTL_TOKEN override
Local admin (init only)	direct DB/PKI bootstrap (no network)	init (before the API is up)	the bootstrap path runs in-process, not over REST

init is special: it runs the bootstrap **before** the API server exists, so it writes state + secrets directly (in-process), not over REST. Every other command is an authenticated REST call [API §6.1].

11.2 Credentials handling (§11.4.10) — load-bearing

```
$HELIXVPCTL_TOKEN          # bearer token (env, never  
logged)                    — highest precedence  
--token <TOKEN>            # explicit flag (discouraged: shell  
history)  
<data-dir>/helixvpctl.token  # file, mode 0600 in a 0700 parent  
dir                        — default source
```

- The token is **never** printed, never logged, never echoed in error messages (§11.4.10). On a 4xx auth error the CLI prints `invalid_token` (the server's code), never the token value.
- The data-dir token file is `chmod 600`, its parent `chmod 700`; the CLI refuses to read a token file with looser perms and prints a remediation hint.
- `enroll-token`'s minted secret and `policy get`'s output go to `stdout`; redirecting to a file is the operator's choice, but `-o json > file` is documented to `chmod 600` the target. The CLI itself writes no secret to any log path.

- A pre-store leak audit (§11.4.10.A) applies if the operator supplies a token to store: grep the tree + history for the value before writing, surface findings.

11.3 mTLS note

REST/operator calls use bearer auth over TLS, NOT device mTLS — device mTLS is the **agent** plane (WatchNetworkMap/AdvertisePrefixes/ReportStatus, [API §6.3]). helixvpctl is an operator tool, not an agent, so it carries an operator API token, never a device cert [API §6.1]. The TLS server cert is verified against the gateway's CA (pinned in the data-dir after init).

12. Global flags & conventions

Flag	Meaning
--control URL / \$HELIXVPCTL_CONTROL	control-plane base URL (default https://localhost:8443)
--token / \$HELIXVPCTL_TOKEN	operator API token (§11.2)
-o, --output table json jsonc	output format (default table for humans, json for scripts)
--ca-file PATH	override the pinned CA (default: data-dir CA)
-v, --verbose	extra diagnostics to stderr — NEVER the token (§11.4.10)

Exit-code convention (uniform): 0 success; 1 domain failure (compile failed, degraded status); 2 usage error (bad flags); 3 missing --confirm on a destructive op; 4 not-found (404); 5 transport/server error (5xx/unreachable); 6 render/IO error. The mapping is stable so scripts can branch on it (the GitOps front door depends on policy set exiting 1 on a bad spec).

13. Test points (§11.4.169 comprehensive test-type coverage)

Every PASS cites captured evidence (§11.4.5/.69/.107); integration drives a real Postgres+Redis booted on demand via vasic-digital/containers (§11.4.76), never ad-hoc docker run.

#	Test type (§11.4.169)	Target	Concrete assertion + evidence
K1	unit	flag parsing + exit-code map (§12)	table-driven: every command's required-flag + exit-code paths
K2	unit	credential never logged (§11.2)	–v output + error paths scanned: token value never appears (mutation injecting it → FAIL)
K3	integration	init → quadlet render → systemctl start (MVP-DoD #1)	fresh init, start the pod, status GREEN; captured transcript
K4	integration	policy set fail- closed (§6)	bad spec → exit 1 + PolicyCompileError, zero policies rows; clean spec → version persisted
K5	integration	enroll-token single-use (§5)	minted token enrolls once; second use → enroll rejected (token consumed, [API §5.4])
K6	integration	device revoke < 1s (§7)	revoke 202 → stream force- close + WG-peer-gone measured < 1s; captured timer
K7	security	token file perms (§11.2)	loose-perm token file → CLI refuses + hints; 0600/0700 enforced
K8	security	no secret in –o json to world- readable file	enroll-token –o json > f documents/sets 0600; mutation removing the perm guard → FAIL
K9	e2e / full- automation	enroll-token → enroll → advertise → policy set -- activate → status	the GitOps loop drives a real reachability change; captured evidence
K10	meta-test (§1.1)	each gate paired	inject token into a log line / skip --confirm guard / persist on bad compile → each makes its gate FAIL

14. Phase-2 forward seams

The `join --control URL --token T` command (a fleet member joining a multi-node control plane) is deferred to Phase 2 [05_OV §5.1] — the single-node `init` path is the MVP. Phase 2 also adds policy GitOps endpoints (policy-as-code with a watched repo), keys rotation scheduling, and a status `--watch` live stream over `/v1/stream` [API §14]. Each is additive; the Phase-1 command tree (§2) draws the verbs in the right shape so Phase-2 growth is new subcommands, not a rename.

Sources verified

- 05-repo-layout-tooling-and-helix-ecosystem.md §5 (helixvpctl command tree, init skeleton
 - artifact set, deploy delegation), §6.3 (substrate render via containers submodule) — [05_OV].
- v03-control-plane/svc-api.md §3 (REST route table), §4 (request/response schemas: enroll-tokens, devices, networks, policies), §4.5 (policy dry-run fail-closed), §5.4 (Enroll single-use), §6.1/§6.3 (authn: API token vs device mTLS), §10.1 (PolicyCompileError) — [API].
- v03-control-plane/protobuf-spec.md §1 (Coordinator messages), §4.5 (AdvertisePrefixes declarative-not-diff), §4.7 (revoke mid-stream < 1s) — [PROTO].
- 04_VPN_CLD/HelixVPN-Phase1-MVP.md §9 (helixvpctl + quadlets, init→systemctl start), HelixVPN-Architecture-Refined.md §4.7 (operator CLI replaces bash scripts) — [04_P1] [04_ARCH].
- Constitution anchors: §11.4.10 (credentials handling — token never logged, file perms, pre-store audit), §11.4.10.A (pre-store leak audit), §11.4.76 (containers submodule for deploy render + test infra), §11.4.81 (cross-platform-parity — one source, three substrates), §11.4.101 (autonomous-decision/reversible — keys rotate --confirm), §11.4.35 (visible reconciliation — verb renames vs §5.1), §11.4.6 (no-guessing — UNVERIFIED exit-code/grace marks), §11.4.169 (test-type coverage), §1.1 (paired meta-test mutations).